



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Y. Sravan Kumar

Reg. No.: 192472289

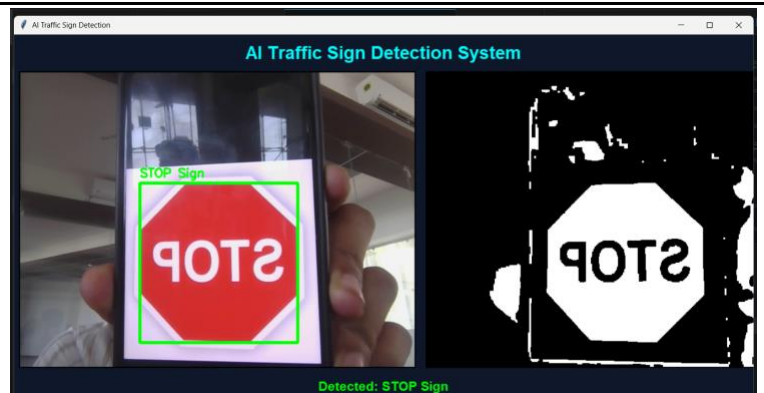
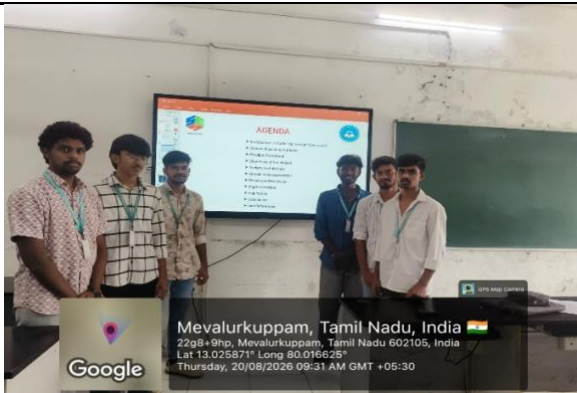
Course Code: ITA0506

Slot: D

Course Name: Computer Vision

Course Faculty: R. C. Jeni Gracia

Project Title: Intelligent Traffic Sign Detection and Recognition using YOLOv8 and CNN with Driver Alert System



Project Description: (here you write what you did in this project (contribution) including Model Description)

Intelligent Traffic Sign Detection and Recognition is an AI-based system developed to improve road safety and assist drivers in real-time. The system uses a camera to capture traffic signs from road scenes and applies to YOLOv8 for fast and accurate detection. A Convolutional Neural Network (CNN) is then used to recognize and classify the detected traffic signs. It can identify signs such as Stop, Speed Limit, No Entry, and Pedestrian Crossing. Once a sign is recognized, the system provides a visual or audio alert to the driver.

This helps drivers respond quickly to important road signs and avoid traffic violations. The system can also reduce driver distraction and improve awareness while driving. Overall, it can be effectively used in Advanced Driver Assistance Systems (ADAS) and future autonomous vehicles.

This module focuses on detecting traffic signs from real-time road images and video using the YOLOv8 object detection algorithm. The system captures video through a camera and processes each frame to identify traffic signs such as Stop, Speed Limit, No Entry, and Warning signs. YOLOv8 analyzes the input frame and identifies the location of each traffic sign using bounding boxes. It also provides a confidence score for every detected sign. The detected signs are displayed on the video with their class names and confidence values. Real-time detection enables the system to recognize traffic signs quickly while the vehicle is moving. This module forms the main detection stage before the detected signs are passed to the CNN classification and driver alert modules.

Student Signature

Guide Signature